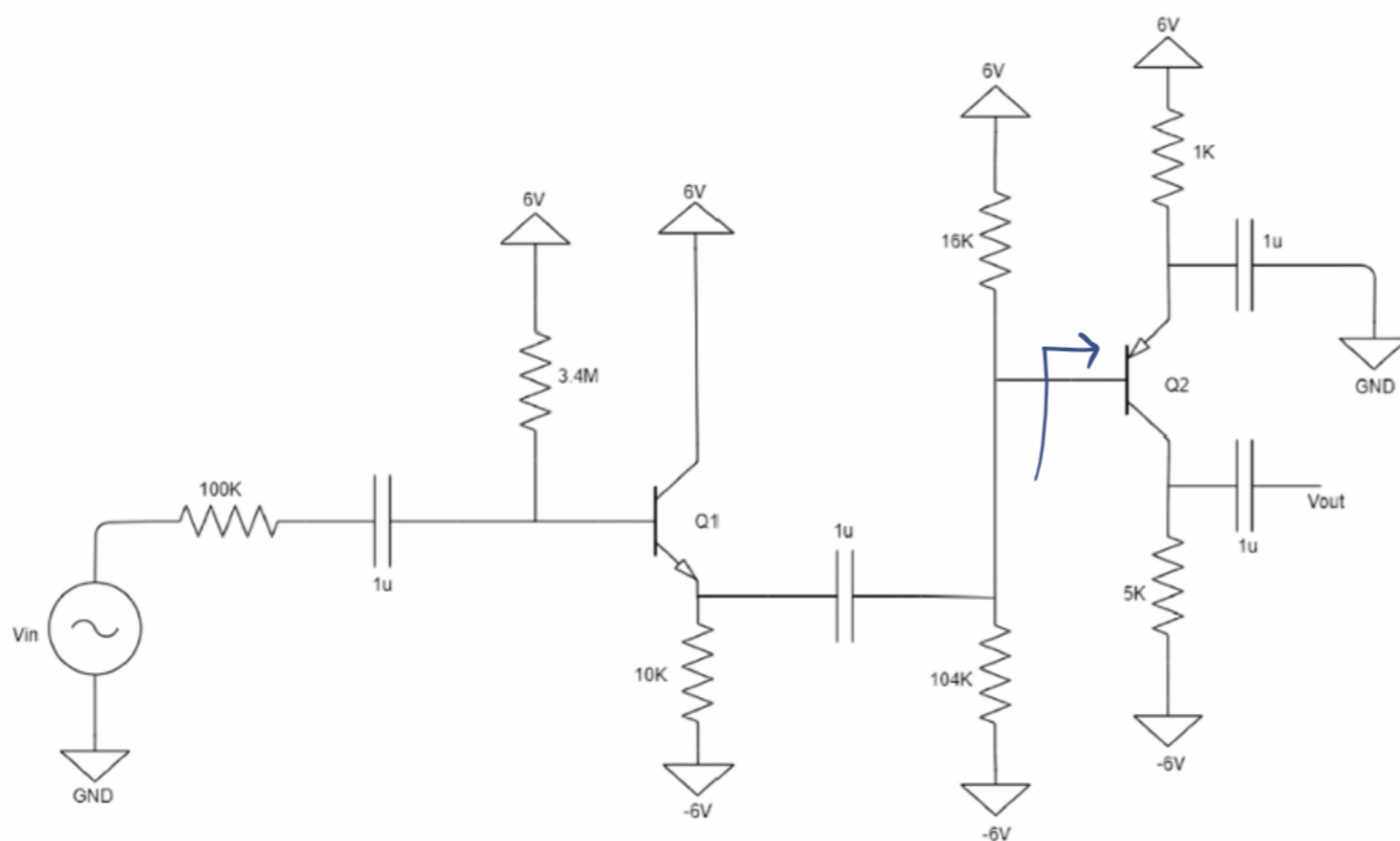
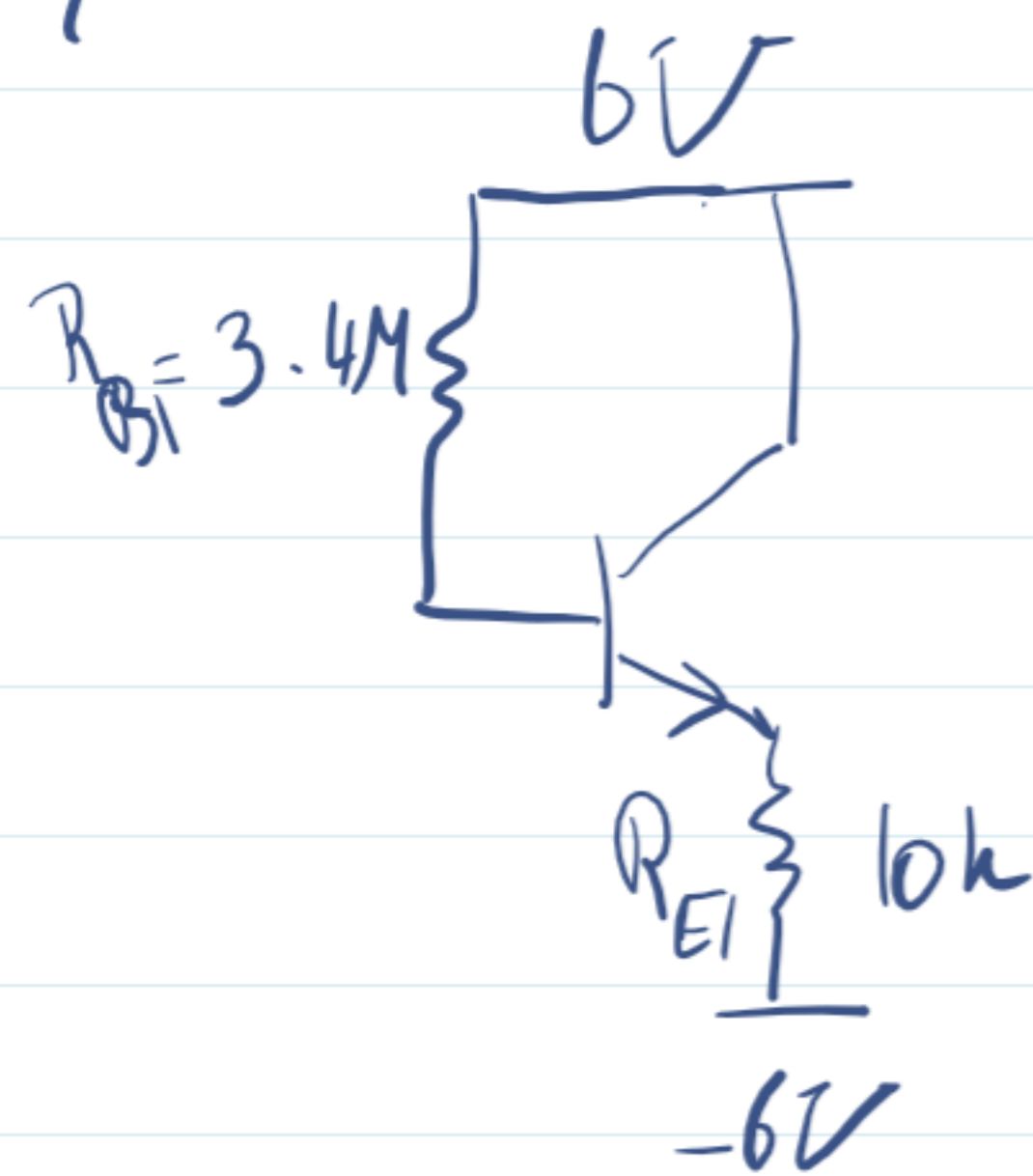




Q1:



$$V_{E1} = -6 + (\beta + 1) \bar{I}_B R_{E1}$$

$$V_{B1} = -5.4 + (\beta + 1) \bar{I}_B R_{E1} = 6 - \bar{I}_B R_{B1}$$

$$\rightarrow \bar{I}_{B1} = \frac{11.4}{R_{B1} + (\beta + 1) R_{E1}} = 2.11 \mu A$$

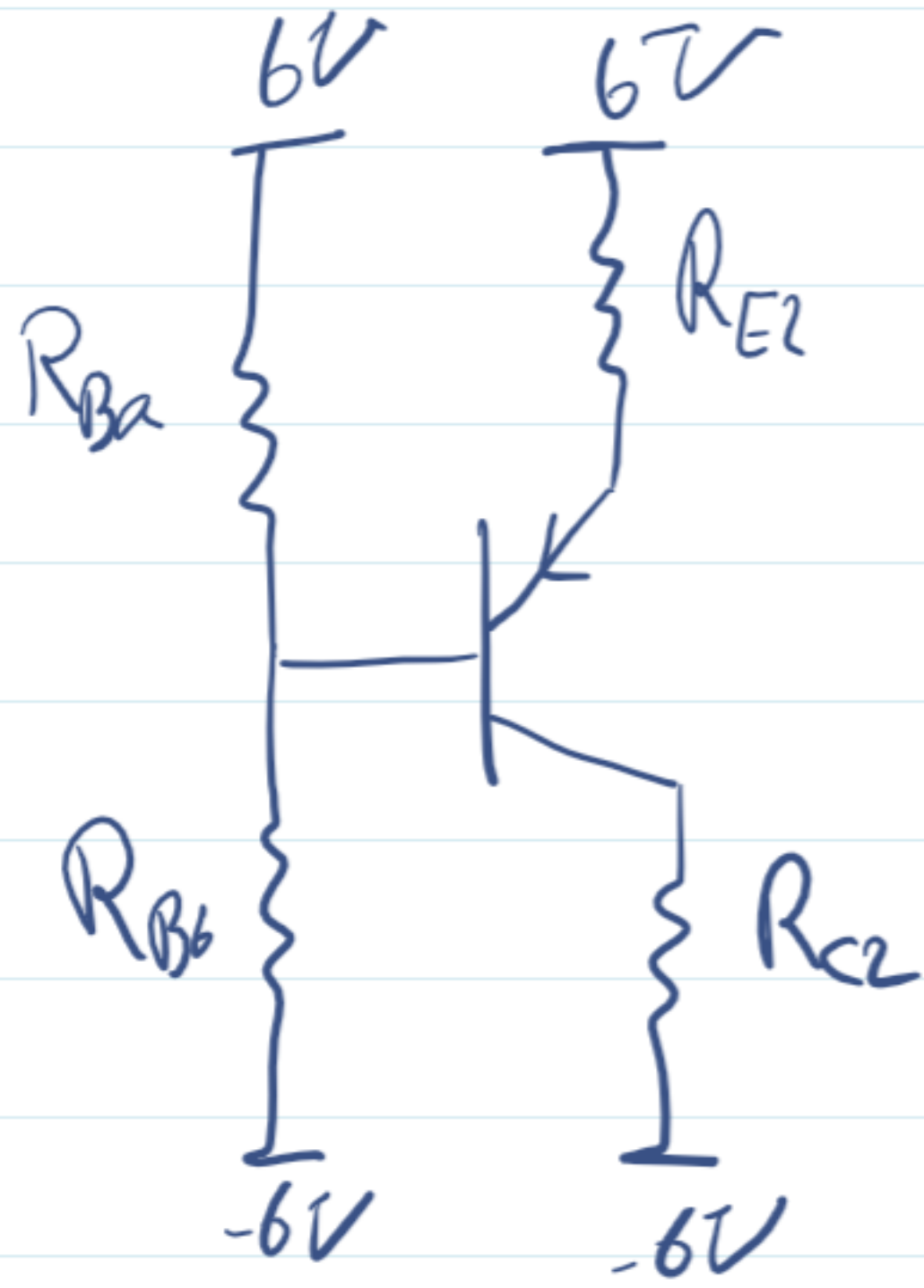
$$\bar{I}_{C1} = 0.421 mA$$

$$V_{E1} = -1.759$$

$$g_m = \frac{\bar{I}_C}{V_T} = 16.8 m$$

$$r_a = \beta / g_m = 11.88 k\Omega$$

Q₂ :



$$V_{B2} = \frac{12 R_{Bb}}{R_{Ba} + R_{Bb}} \cdot 6V = 4.4V$$

$$V_{E2} = V_{B2} + V_{BE(on)} = 5V$$

$$I_C = 1mA \quad V_C = -1$$

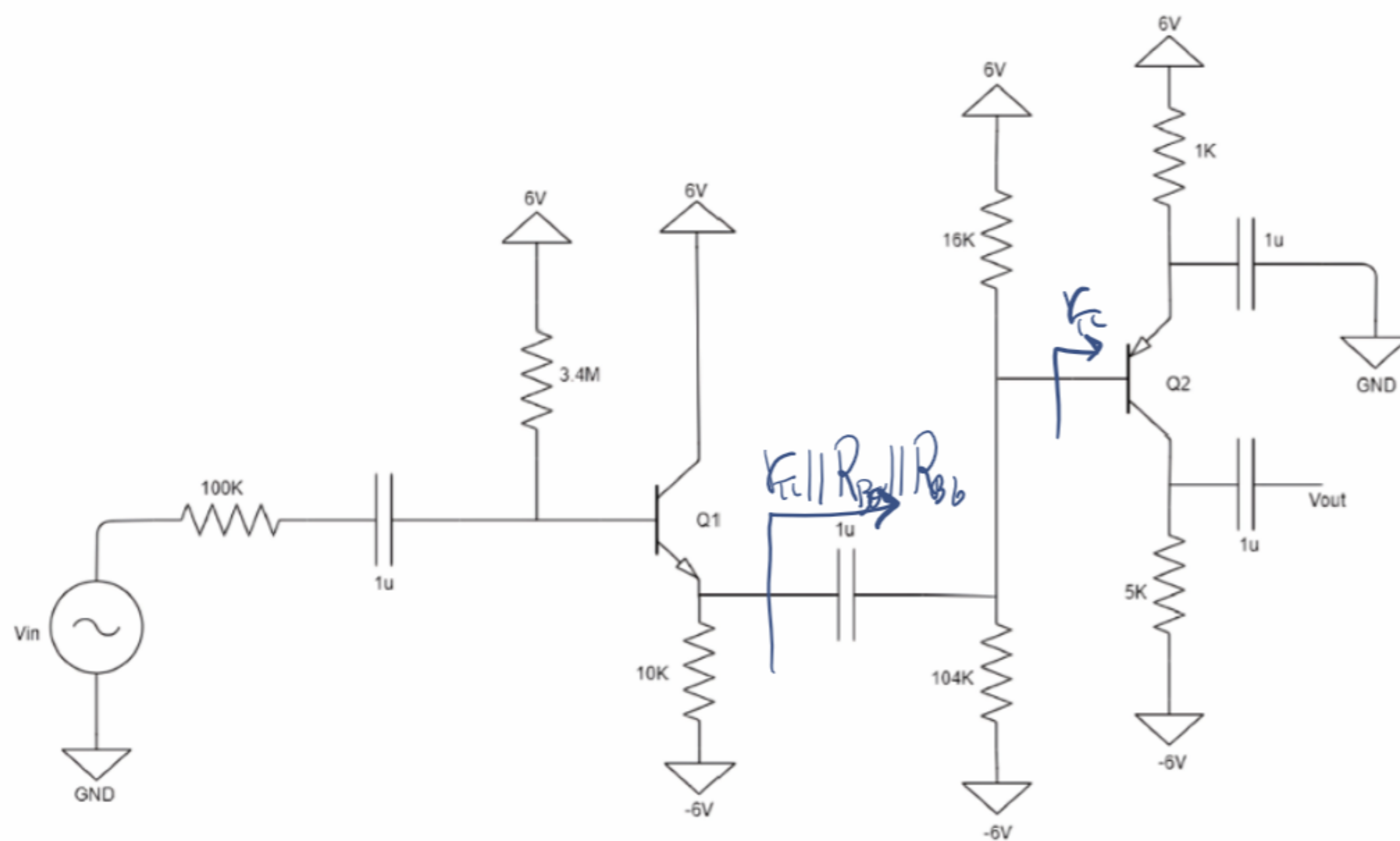
$$g_m = 40mS^{-1}$$

$$r_a = 2.5k\Omega$$

	Q ₁	Q ₂
I_C	0.421mA	1mA
g_m	16.8 mS ⁻¹	40mS ⁻¹
r_a	11.88 kΩ	2.5kΩ

$$A_{Q2} = -g_{m2} R_{C2}$$

(-)



$$R_{E_{eq}} = R_{E1} \parallel r_{E2} \parallel R_{B1} \parallel R_{B2} = 1748 \Omega$$

$$A_{Q1} = \frac{R_{E_{eq}}}{R_{E_{eq}} + \frac{R_s + r_e}{\beta_1}} = 0.76$$

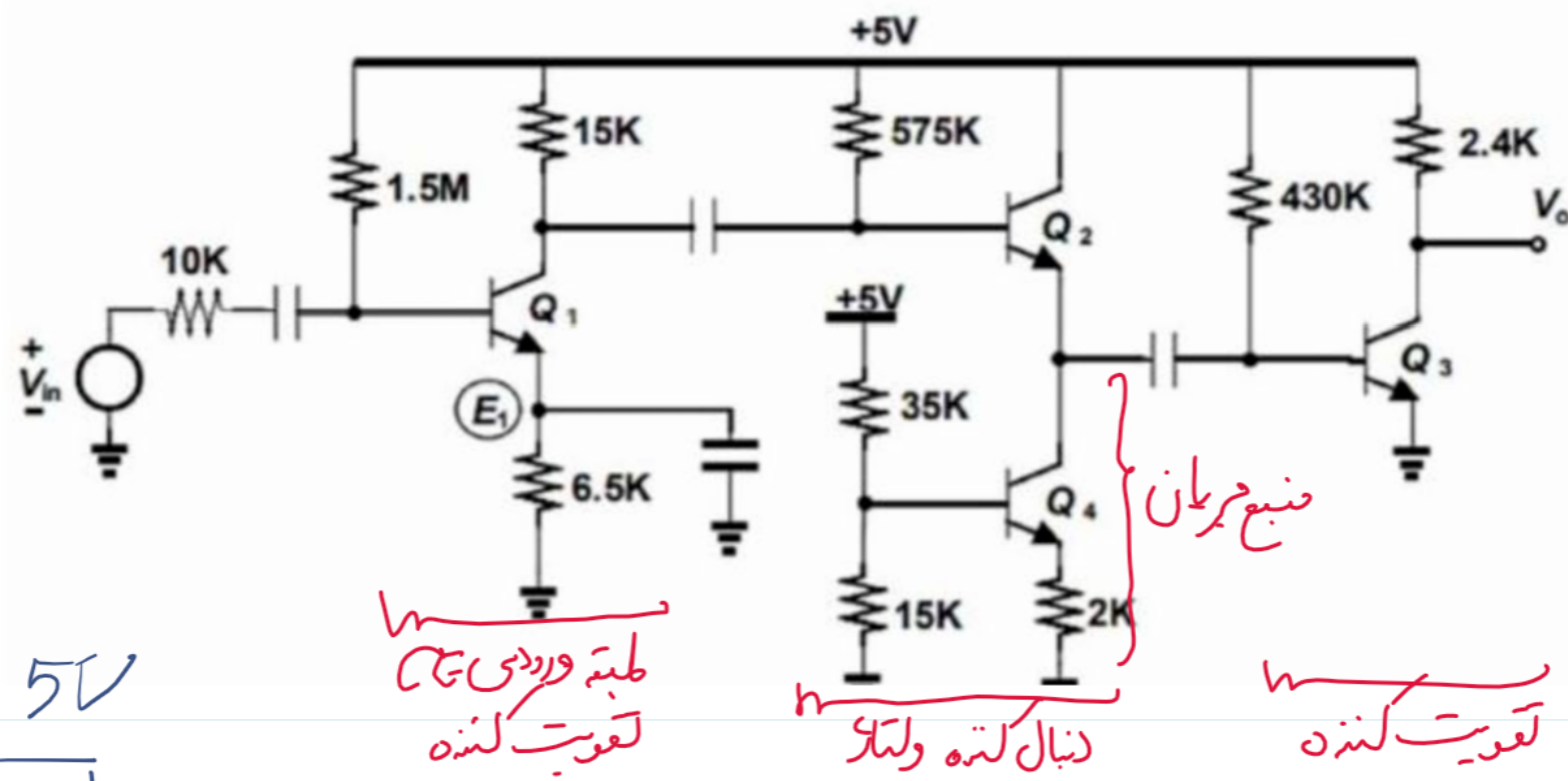
$$A_{Q2} = -200$$

$$A_v = -152$$

17

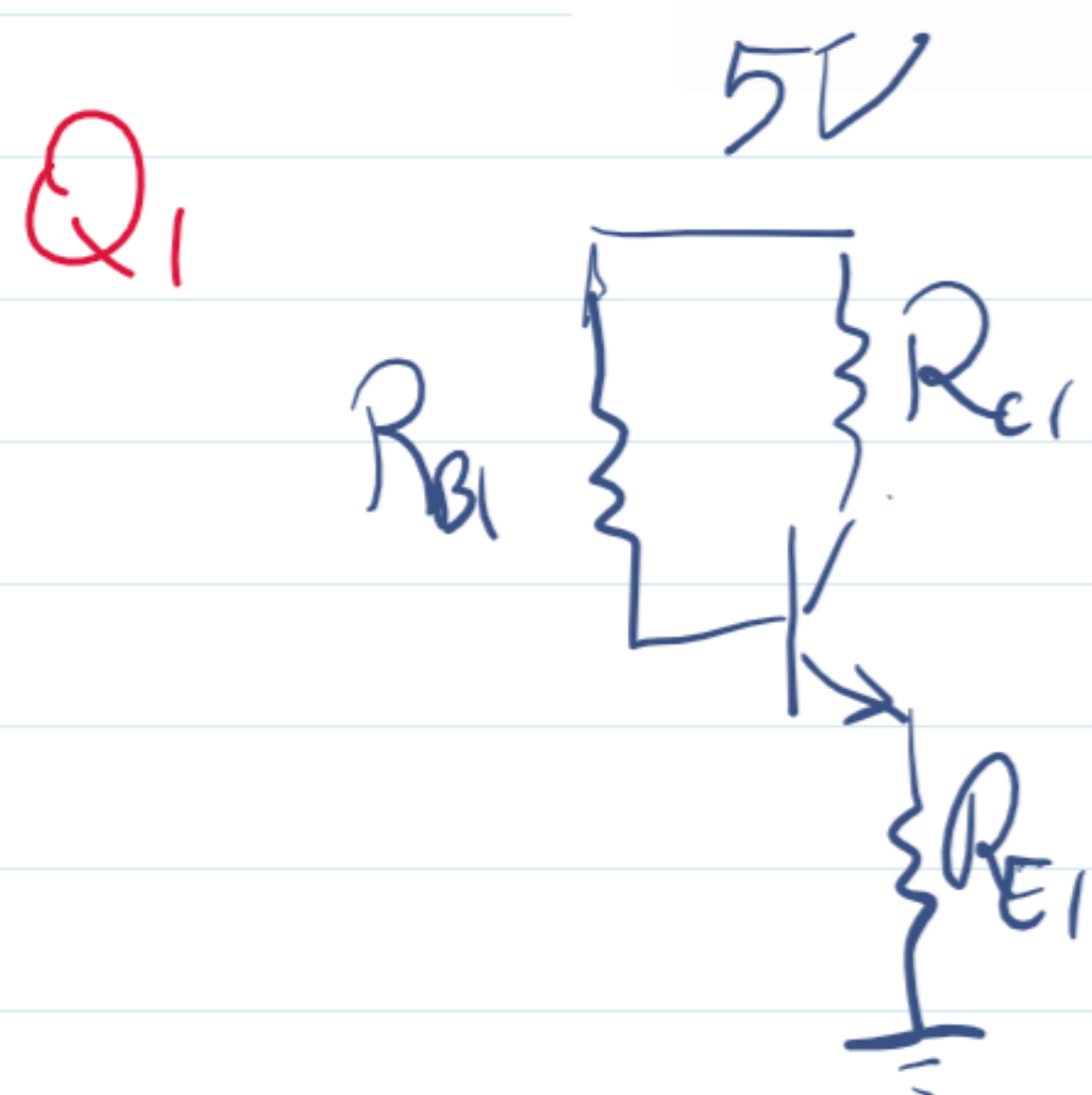
$$R_{out} = R_{oQ2} = 5 k\Omega$$

$$R_{in} = \beta R_{E_{eq}} + r_e = 350 k\Omega$$



$$\beta = 100$$

$$V_{BE(on)} = 0.7V$$



$$V_E = (\beta + 1) I_B R_{E1}$$

$$V_B = V_{BE(on)} + (\beta + 1) I_B R_{E1} = 5 - I_B R_{B1}$$

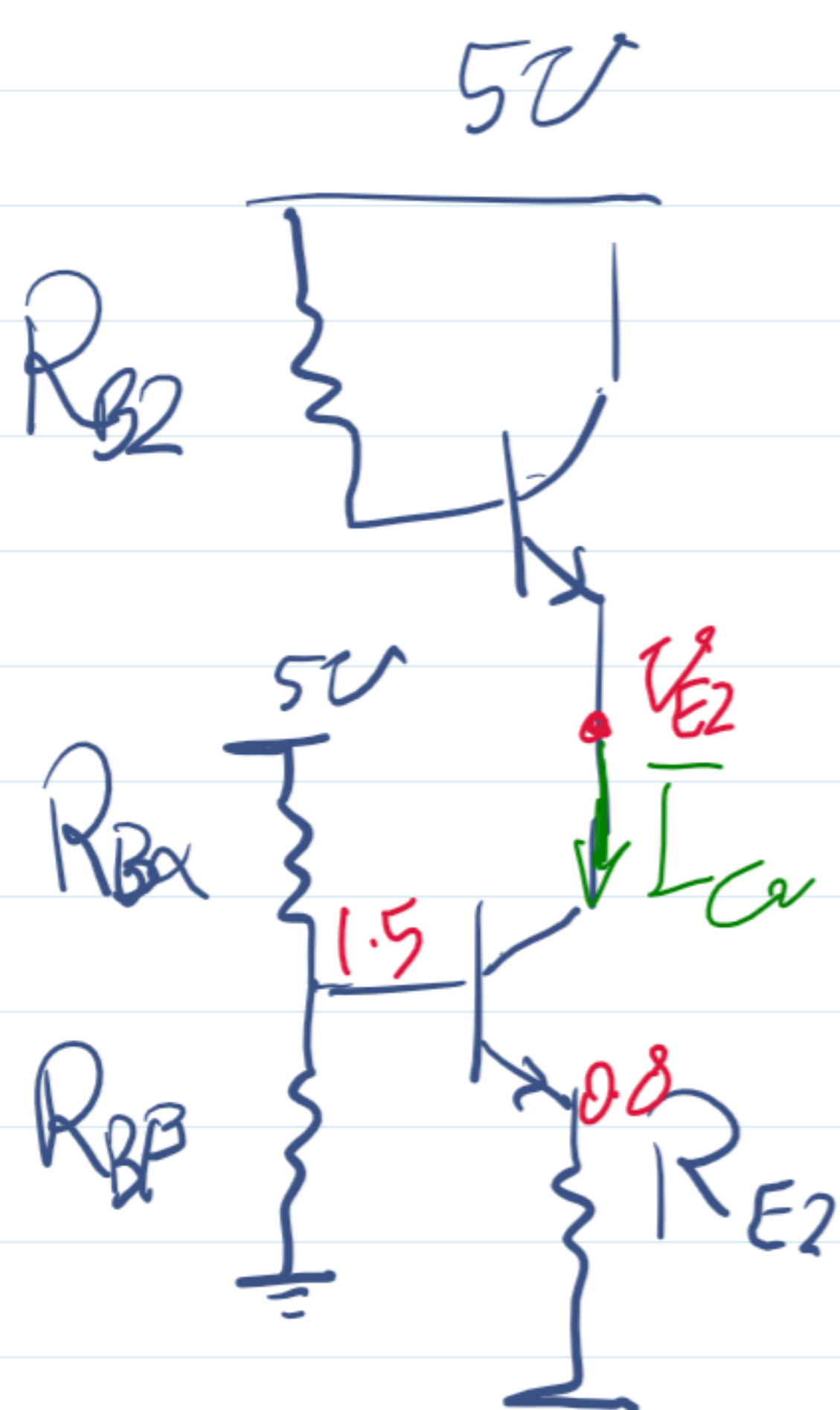
$$\rightarrow I_{C1} = \frac{4.3 \beta}{(\beta + 1) R_{E1} + R_{B1}} = 0.2 mA$$

$$V_C = 2V \quad V_E = 1.3V$$

$$r_c = 12.5 k\Omega$$

$$g_m = 8 m\Omega^{-1}$$

Q2:



$$V_{BQ2} = \frac{R_{B2} \beta}{R_{B2} + R_{B1}} 5V = 1.5V$$

$$I_{C2} = \frac{0.8V}{R_{E2}} = 0.4 mA$$

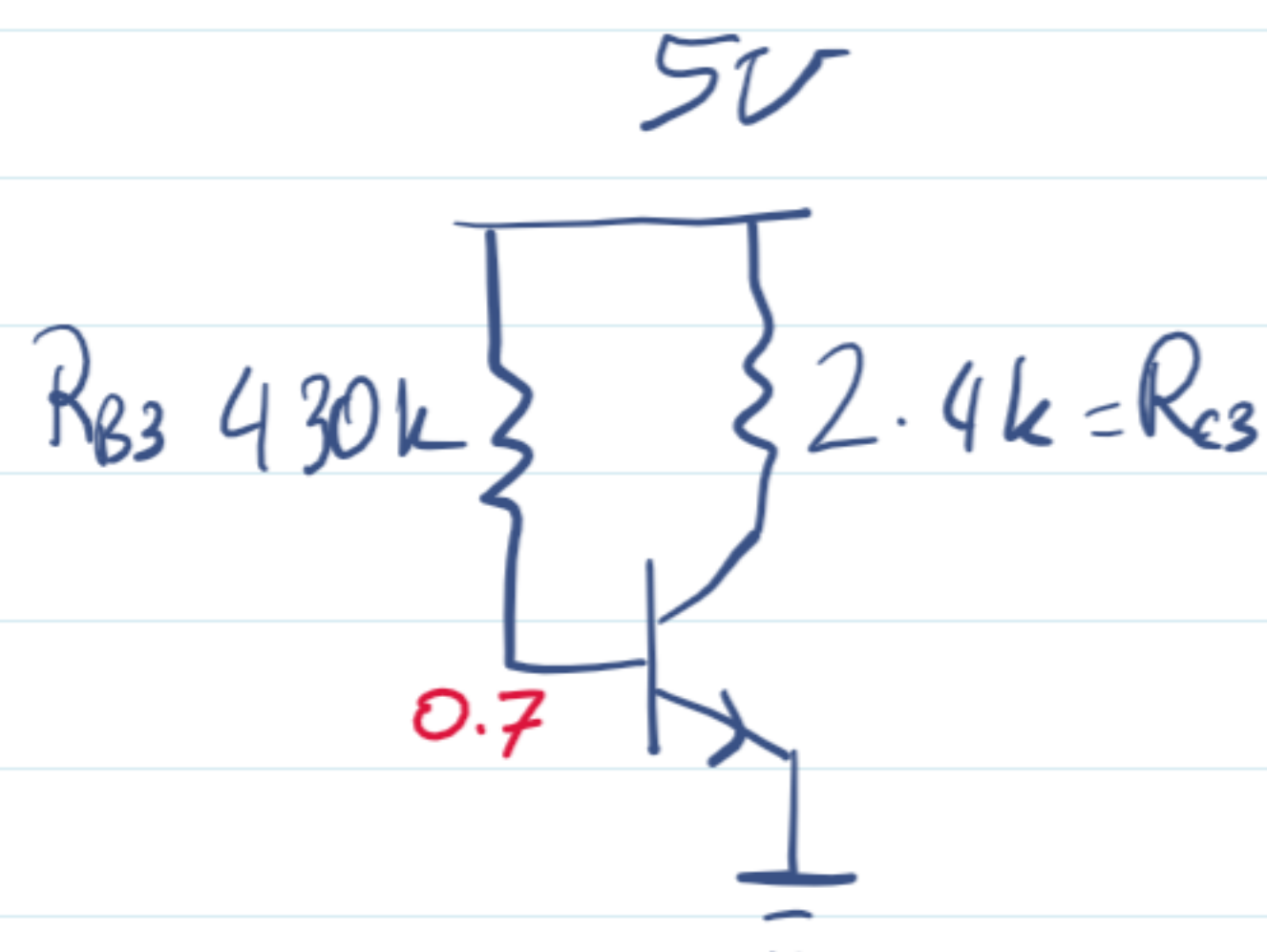
$$V_{B2} = 5 - R_{B2} I_{C2} = 2.7V$$

$$g_m = 16 m\Omega^{-1}$$

$$V_{E2} = V_{B2} - V_{BE(on)} = 2V$$

$$r_a = 6.25 k\Omega$$

Q3:

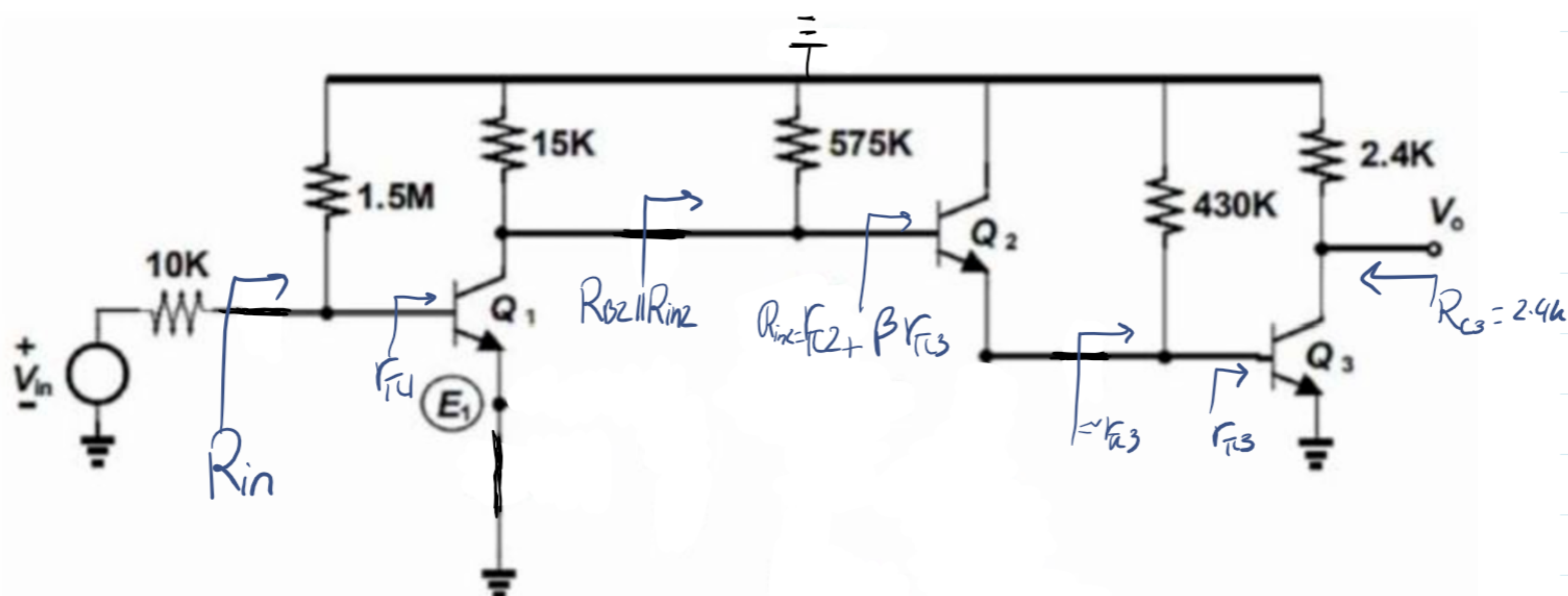


$$I_{C3} = \beta \frac{5 - V_{BE(on)}}{R_{B3}} = 1 \text{ mA}$$

$$V_C = 2.6 \text{ V}$$

$$g_m = 40 \text{ m}\Omega^{-1}$$

$$r_{\pi} = 2.5 \text{ k}\Omega$$



$$R_{in} = 1.5 \text{ M} \parallel r_{\pi 1} \approx r_{\pi 1} = 12.5 \text{ k}\Omega$$

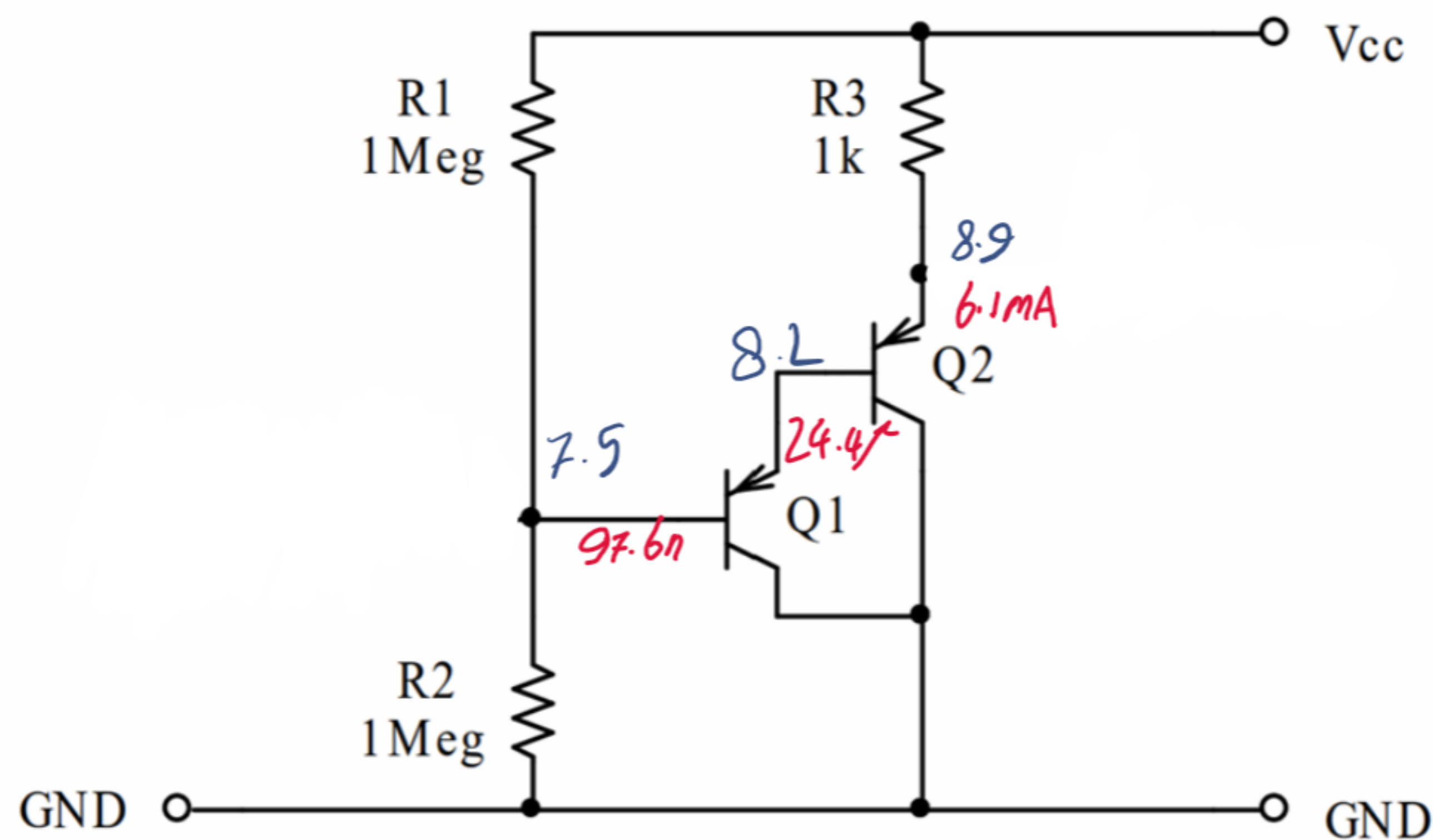
$$R_{out} = R_{C3} = 2.4 \text{ k}\Omega$$

$$A_{Q1} = -g_{m1} R_{C,eq} = -g_{m1} (15 \text{ k} \parallel 575 \text{ k} \parallel (r_{\pi 2} + \beta r_{\pi 3})) = -110.64$$

$$A_{Q2} = \frac{R_{E,eq}}{R_{E,eq} + \frac{R_S + r_{\pi 2}}{\beta}} \approx \frac{r_{\pi 3}}{r_{\pi 3} + \frac{R_{C1} + r_{\pi 2}}{\beta}} \approx 0.92$$

$$A_{Q3} = -g_{m3} R_{C3} = -96$$

$$\rightarrow A \approx 9800$$



جریان: A
ولتاژ

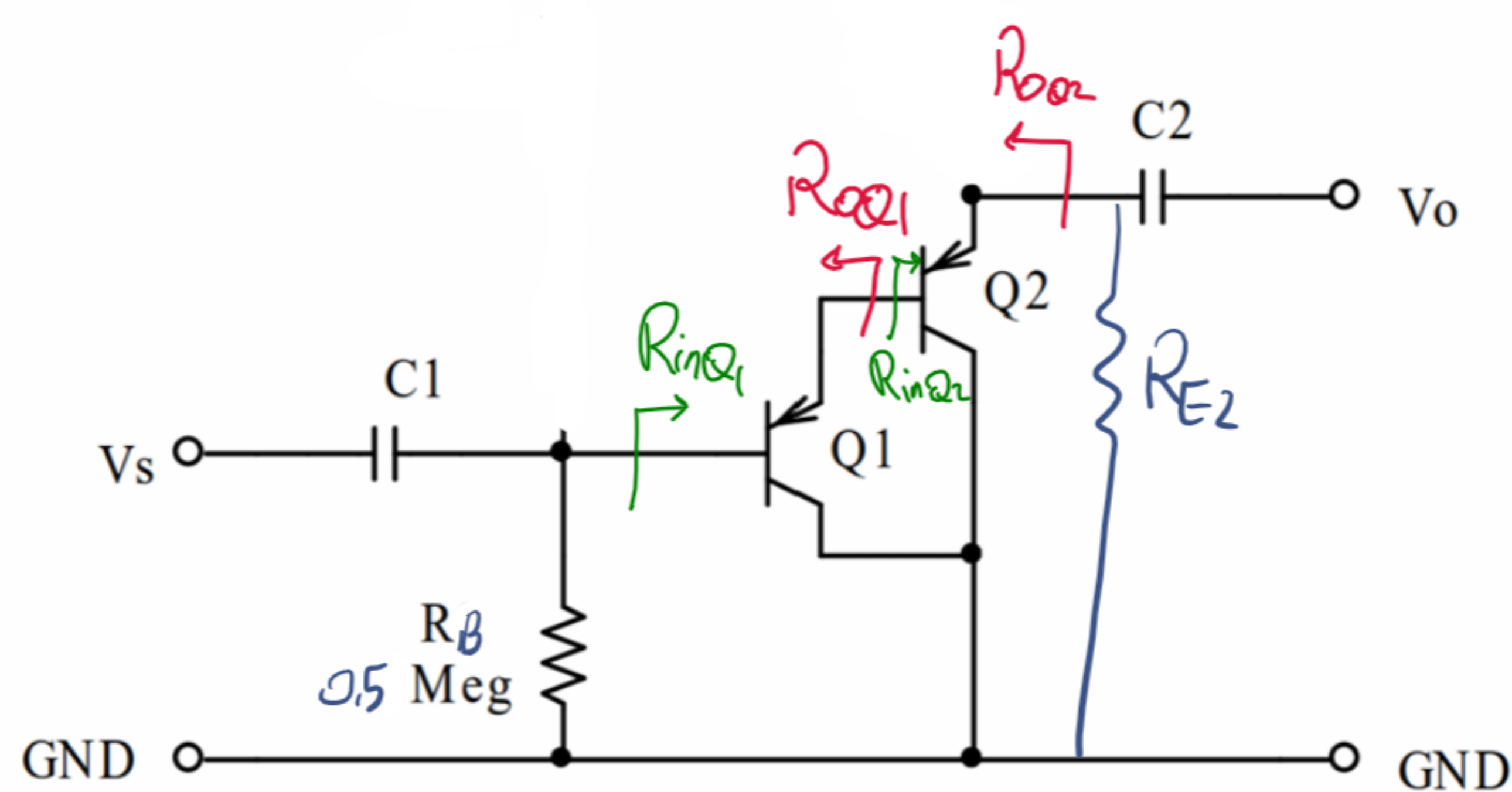
شرط تأثیر نداشتن :
جریان بیس برابر است
 $97.6 \mu A \ll \frac{15V}{2M}$

$$g_{m1} = 0.976 \text{ m}\Omega^{-1}$$

$$g_{m2} = 244 \text{ m}\Omega^{-1}$$

$$r_{\pi1} = 256 \text{ k}\Omega$$

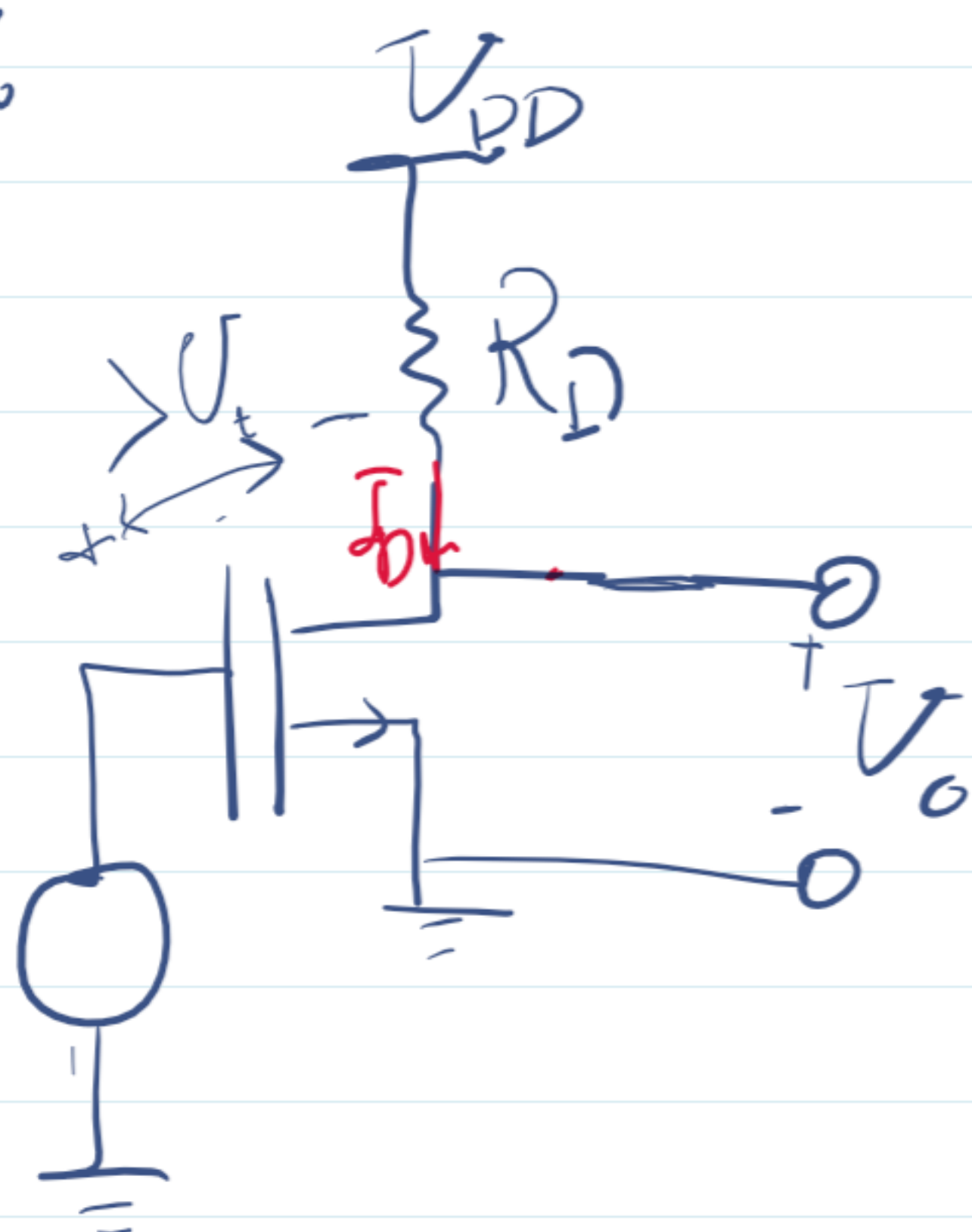
$$r_{\pi2} = 1025 \Omega$$



$$R_{oQ1} = \frac{R_B + r_{\pi1}}{\beta} = 3024 \Omega$$

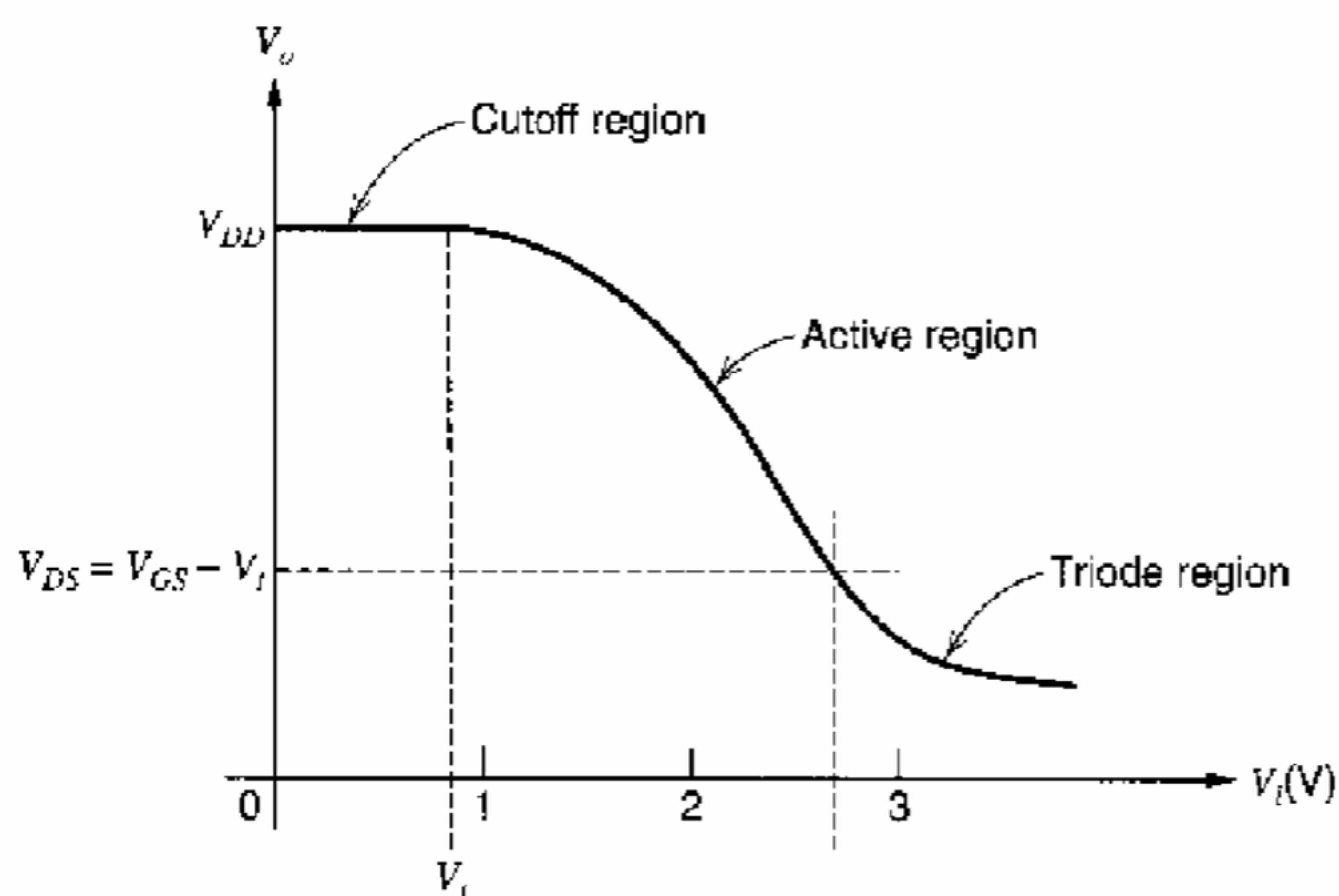
$$R_{oQ2} = \frac{R_{oQ1} + r_{\pi2}}{\beta} = \frac{r_{\pi2}}{\beta} + \frac{r_{\pi1} + R_B}{\beta^2} = 16.2 \Omega$$

CS:



$$V_o = V_{DD} - I_D R_D$$

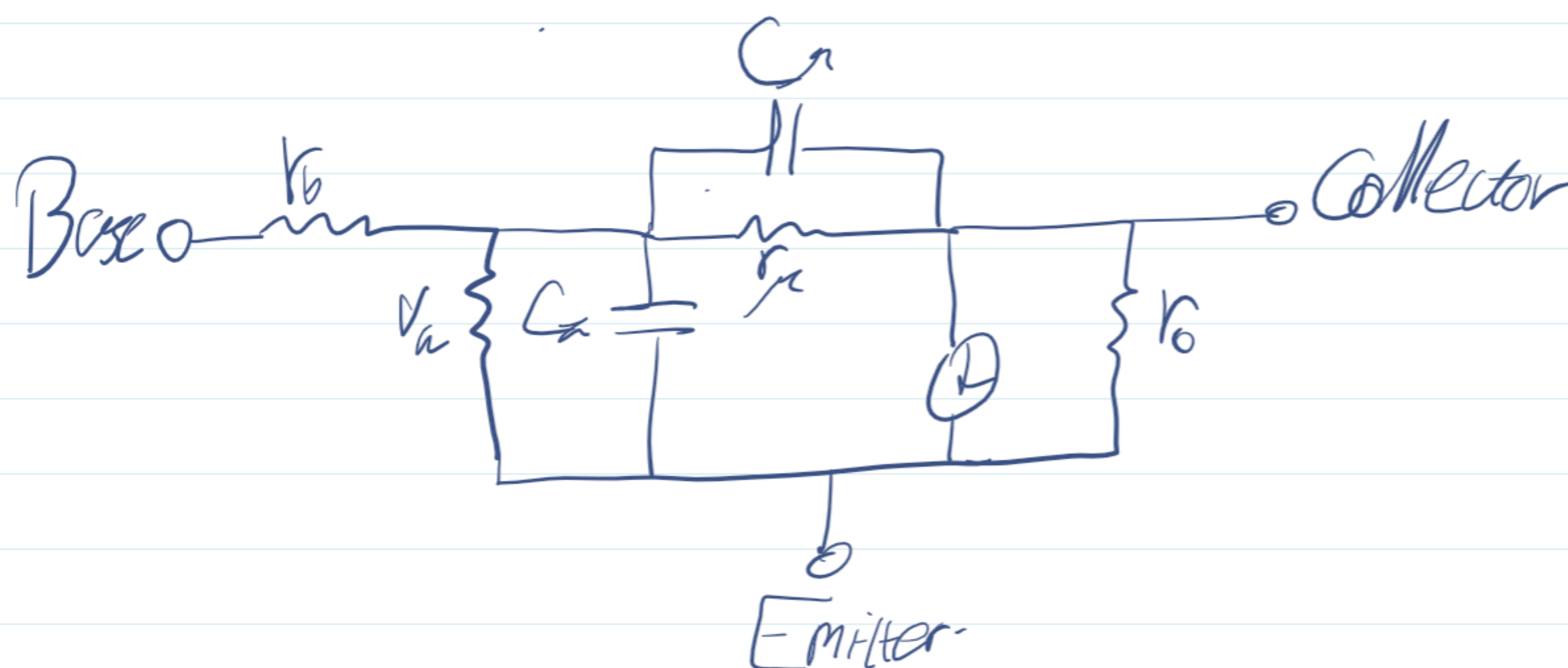
$$V_{GD} > V_t$$

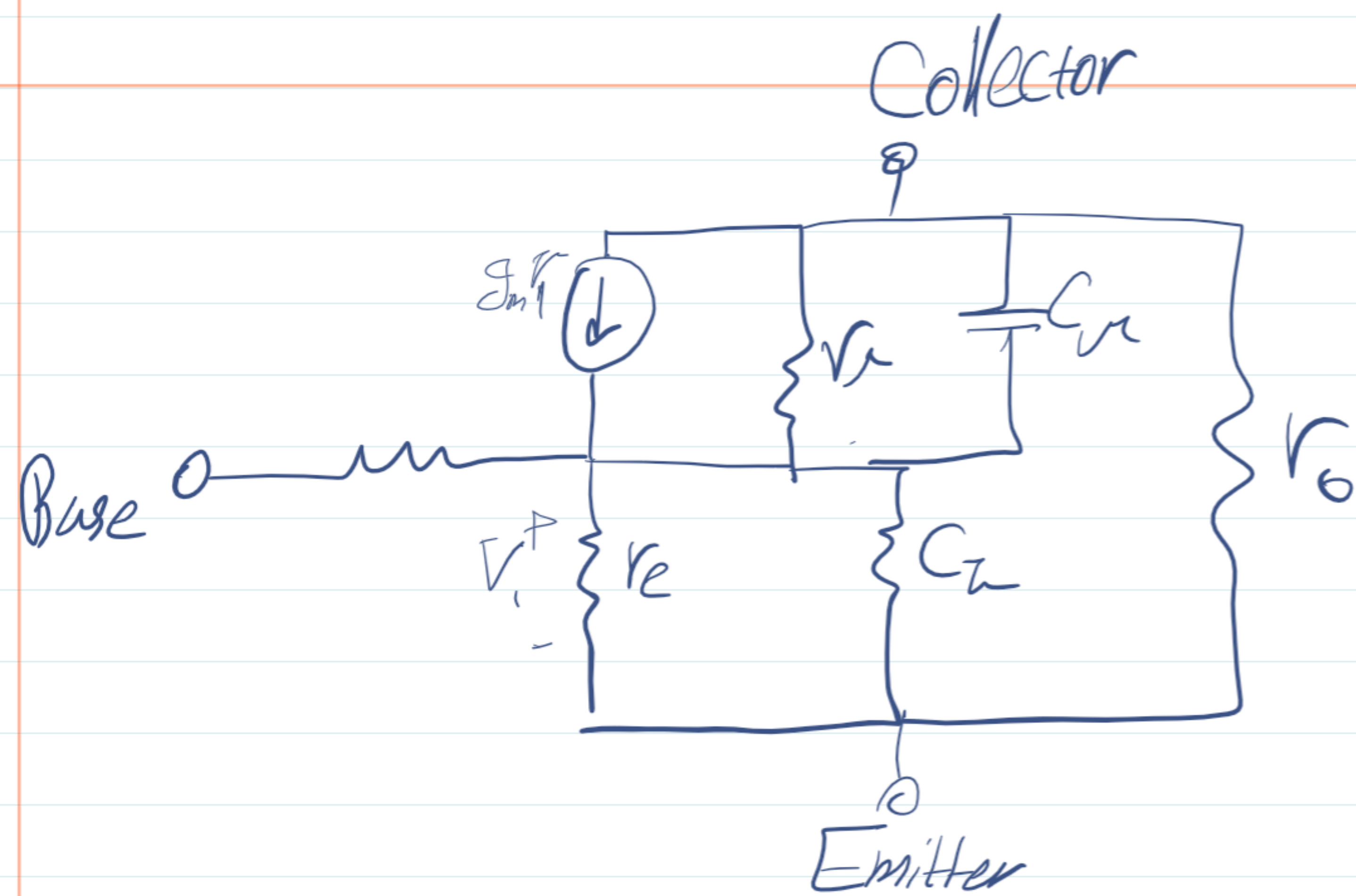


$$g_m = 2\sqrt{k I_D}$$

$$r_o = \frac{V_A + V_{DS}}{I_D}$$

in common-base stages the hybrid- π model could be complicated.

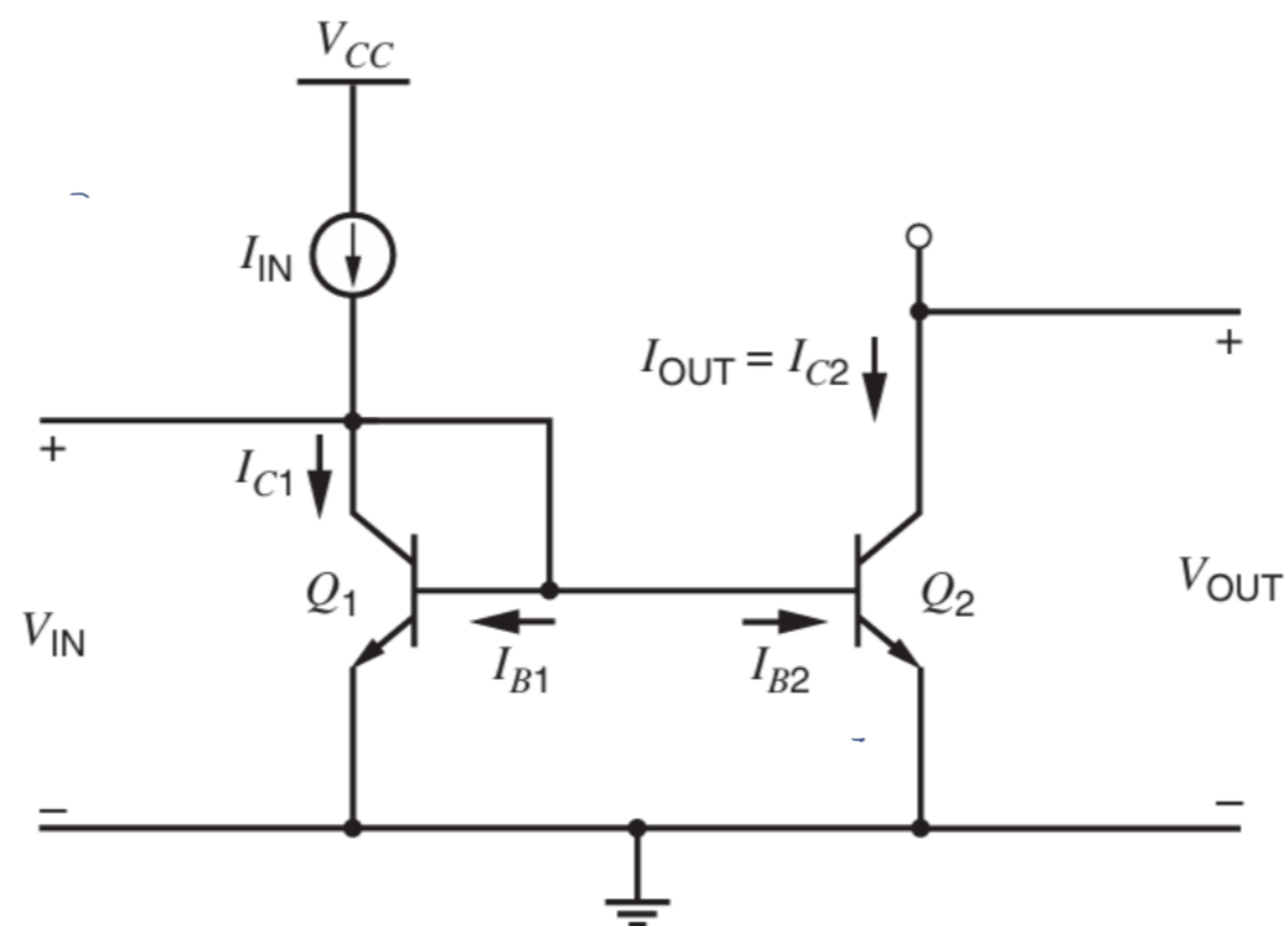




Which r_e is:

$$r_e = \frac{1}{g_m + 1/r_{\pi}}$$

Current mirror:

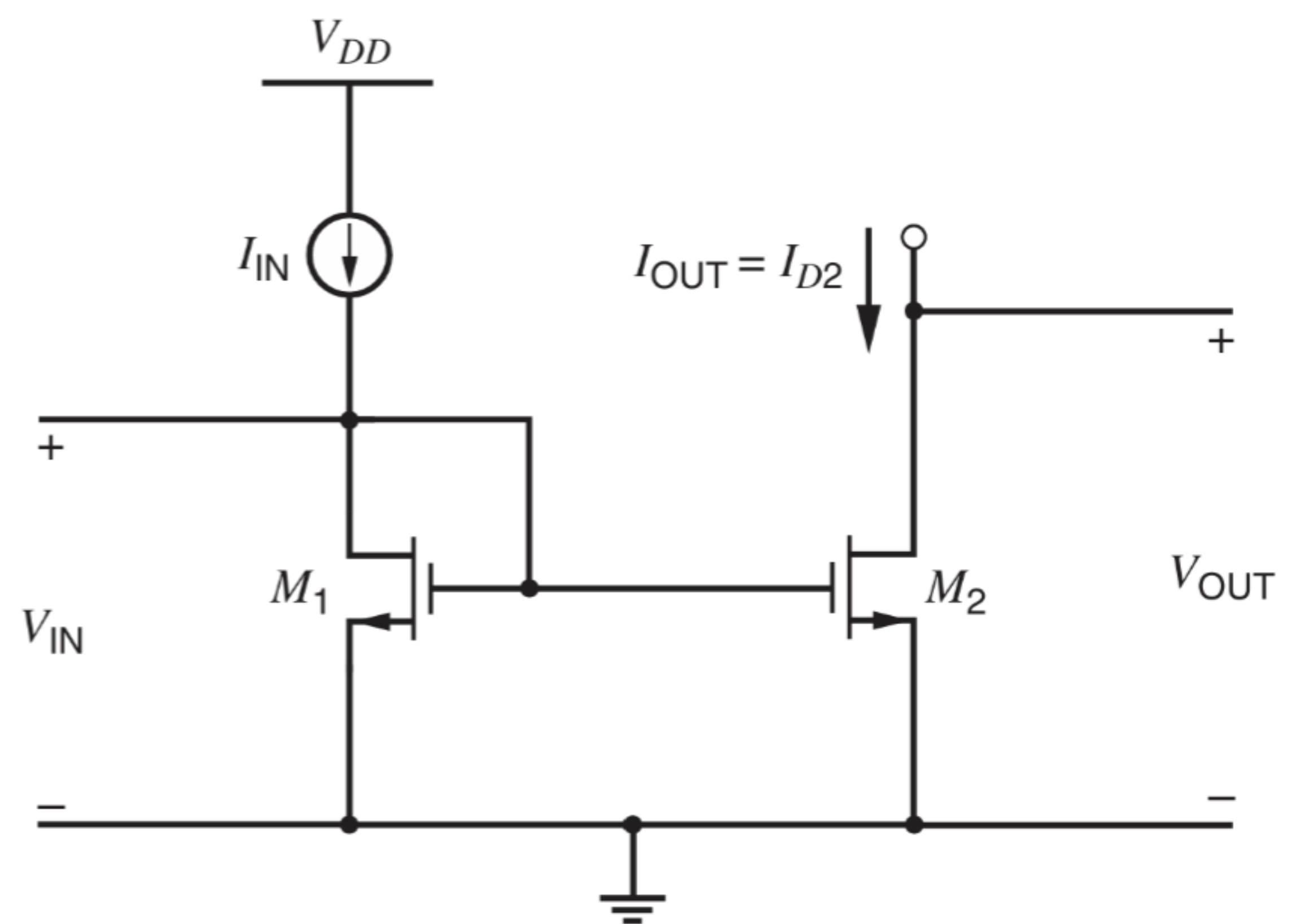


Simple current mirror with BJT.

MOS:

$$I_D = \frac{1}{2} k \frac{W}{L} (V_{GS} - V_t)^2$$

If two transistor be identical
 I_{out} would be equal to I_{in}



$$R_{inQ2} = r_{\pi 2} + \beta R_{E2} \approx 250 \text{ k}\Omega$$

$$R_{inQ1} = r_{\pi 1} + \beta R_{inQ2} = 62.5 \text{ M}\Omega$$

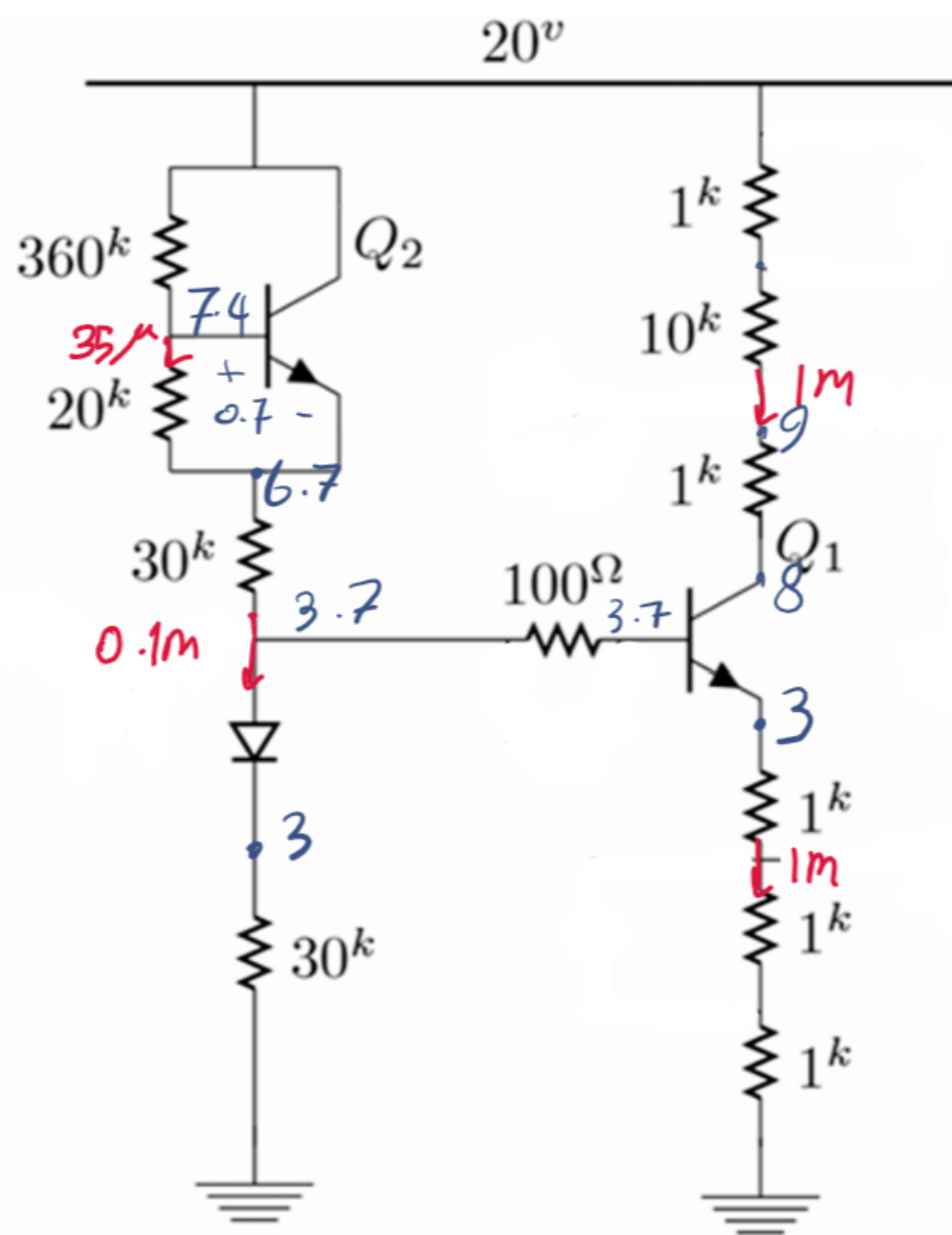
$$A_{Q2} = \frac{R_{E2}}{R_{E2} + \frac{r_{\pi 2} + R_{inQ1}}{\beta}} \approx 1$$

$$A_{Q1} = \frac{R_{inQ1}}{R_{inQ1} + \frac{r_{\pi 1} + R_B}{\beta}} \approx 1$$

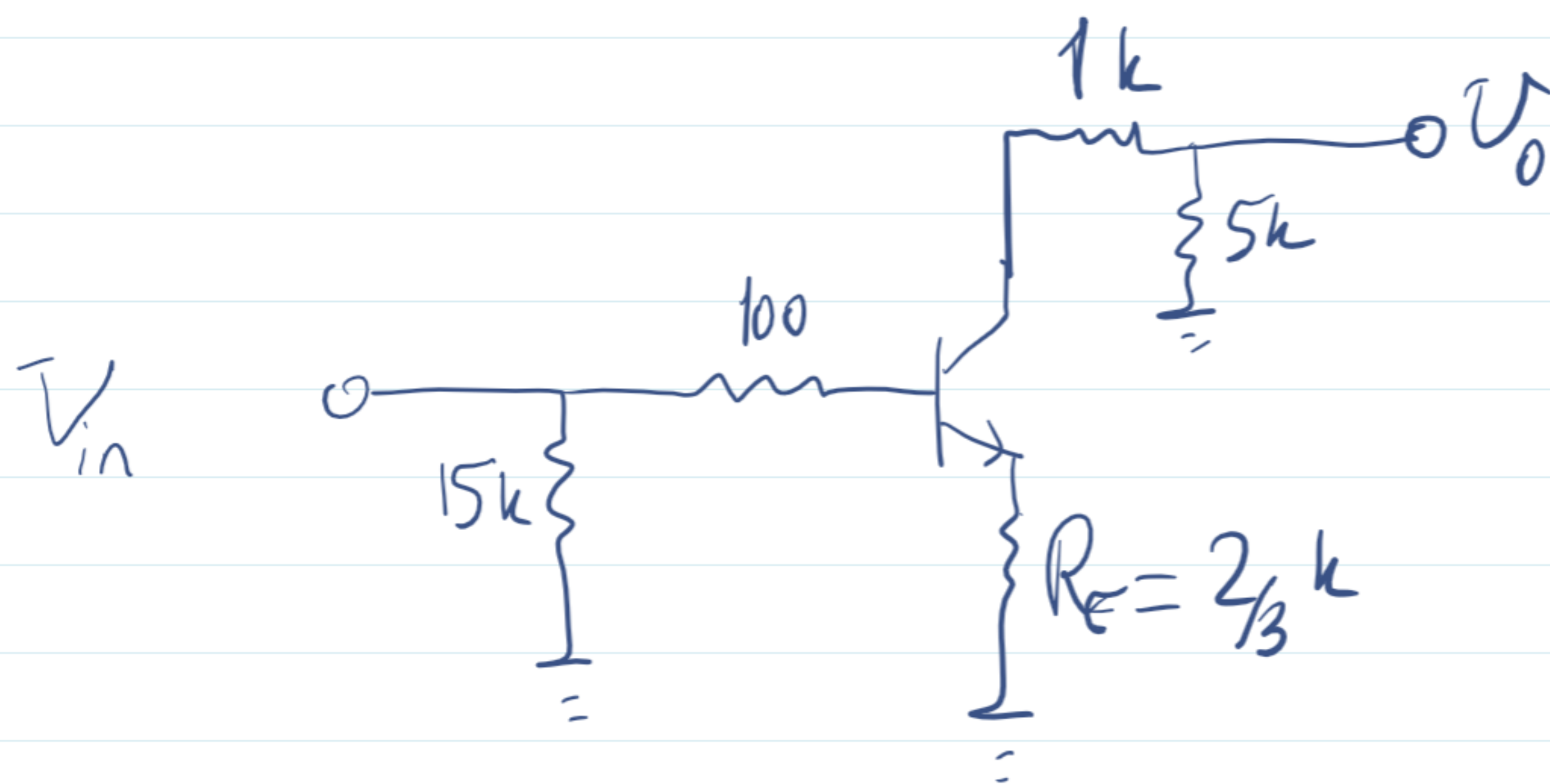
$$\rightarrow A \approx 1$$

$$R_{in} = 0.5 \text{ M} \parallel 62.5 \text{ M} = 0.5 \text{ M}\Omega$$

$$R_{out} = 16.2 \text{ }\Omega$$



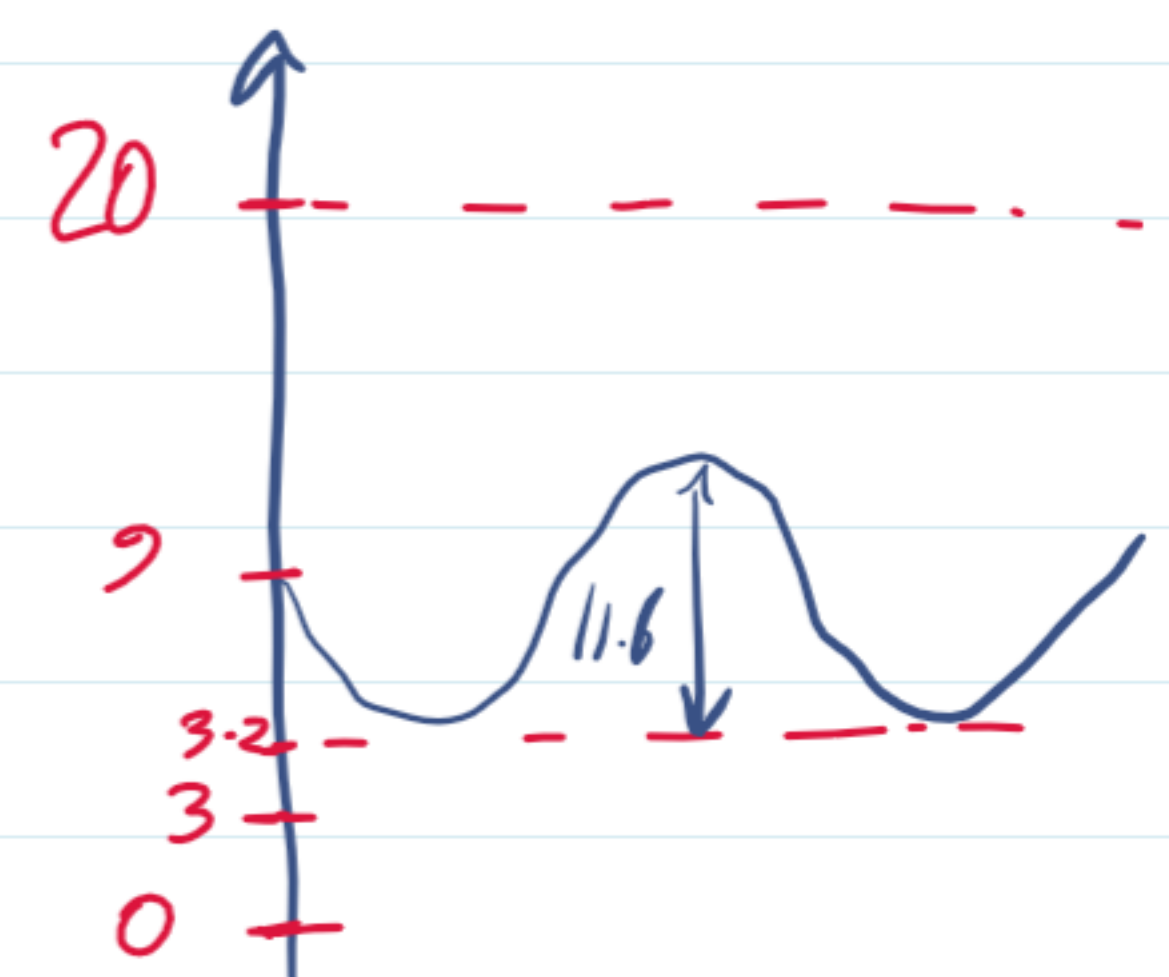
فرض اولیه: جریان بیس Q_2 نسبت به $35\mu A$ بسیار کوچک است. و هیستود جریان بیس Q_1 فرض اولیه حالت بار بزرگ تا خطا در محاسباتی نشود.

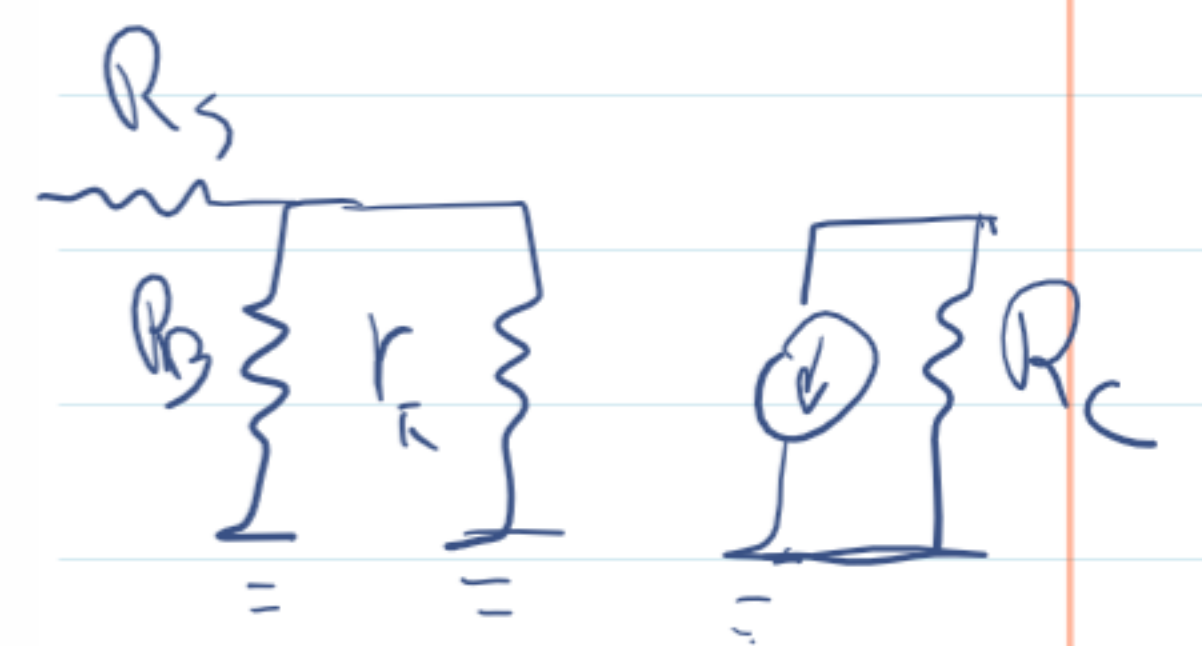
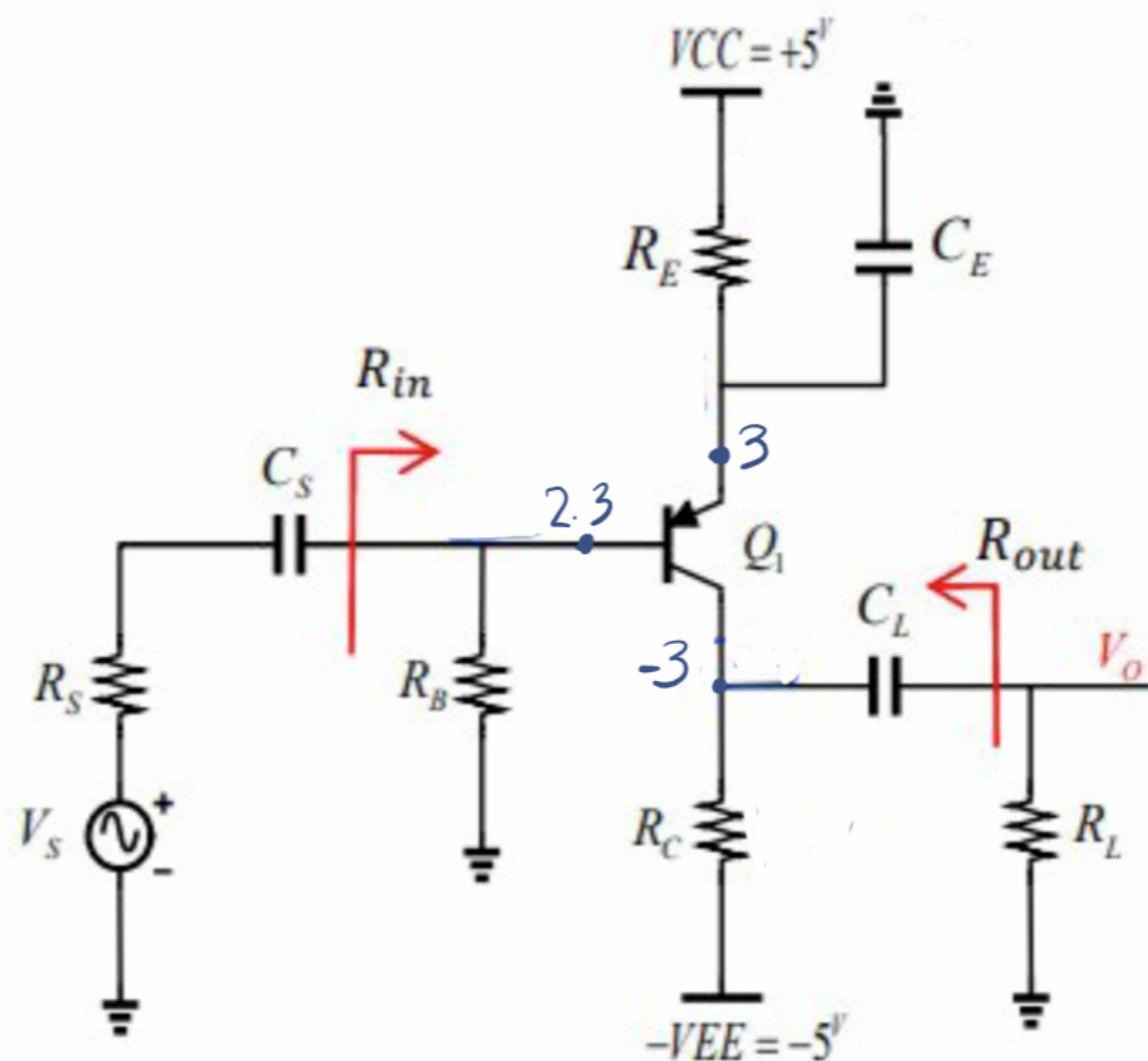


$$R_o = 5k$$

$$R_{in} = 15k \parallel (100 + \underbrace{2.5k}_{r_{\pi}} + \beta R_E) = 12.33k$$

$$A = -g_m (6k) = -240$$





$$V_E = V_{CC} - (\beta + 1) \bar{I}_B R_E$$

الف

$$\bar{I}_B = \frac{V_E - 0.7}{R_B} = \frac{4.3 - (\beta + 1) \bar{I}_B R_E}{R_B} \Rightarrow \bar{I}_B = \frac{4.3}{R_B + (\beta + 1) R_E}$$

$$\bar{I}_C = \frac{\beta}{R_B + (\beta + 1) R_E} (4.3) = 0.5 \text{ mA}$$

$$g_m = 20 \text{ m}\Omega^{-1}$$

$$r_\pi = 5 \text{ k}\Omega$$

$$R_{in} = r_\pi \parallel R_B = r_\pi = 5 \text{ k}\Omega$$

$$R_{out} = R_C = 4 \text{ k}\Omega$$

$$\frac{V_B}{V_S} = \frac{R_{in}}{R_S + R_{in}}$$

$$\frac{V_{out}}{V_B} = -g_m R_C$$

$$\frac{V_{out}}{V_S} = \frac{-g_m R_{in} R_C}{R_S + R_{in}} = -78.43$$

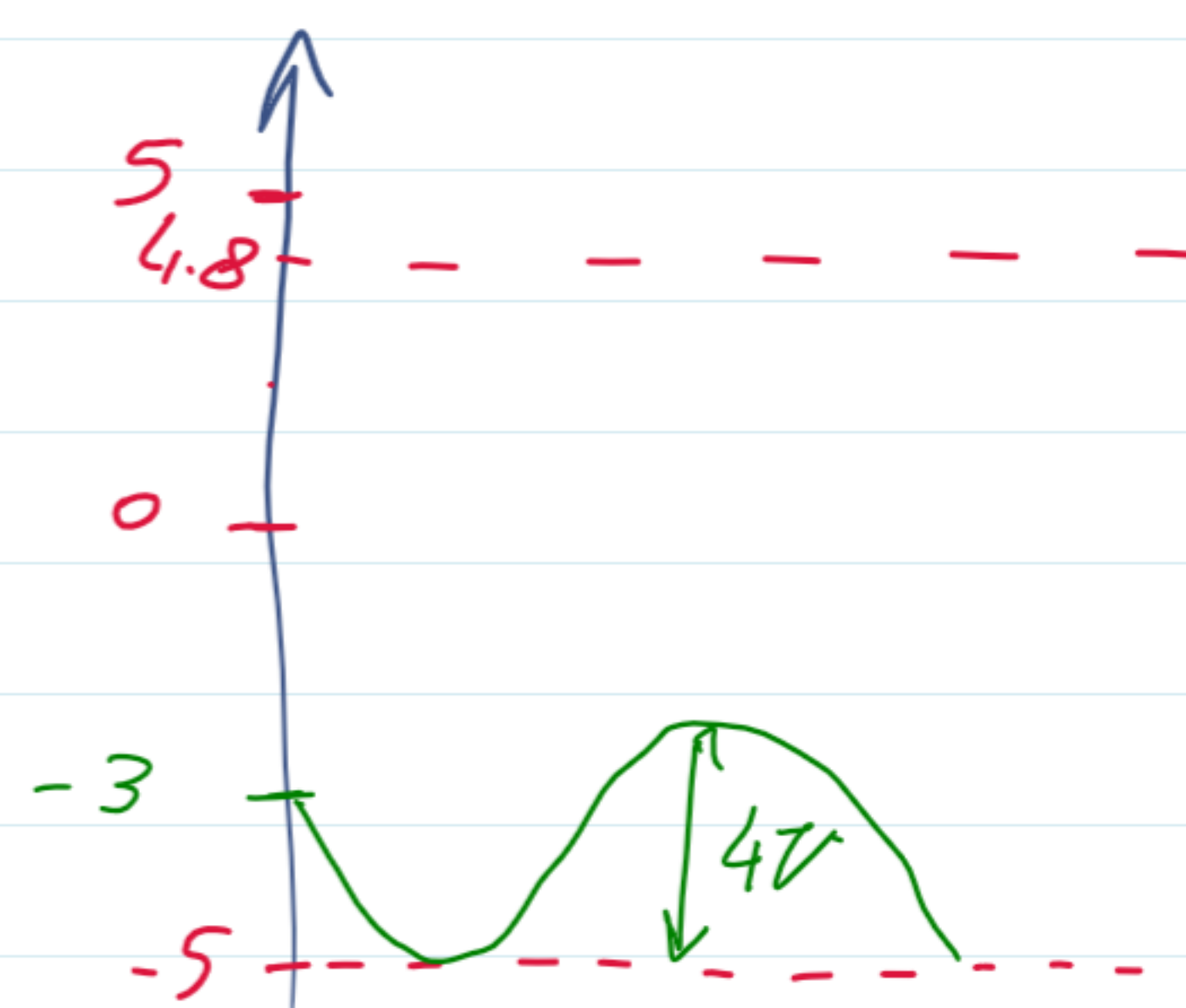
(ب)

$$R_L = \infty$$

$$V_{O+out} > -5 \rightarrow V_{out} > -2$$

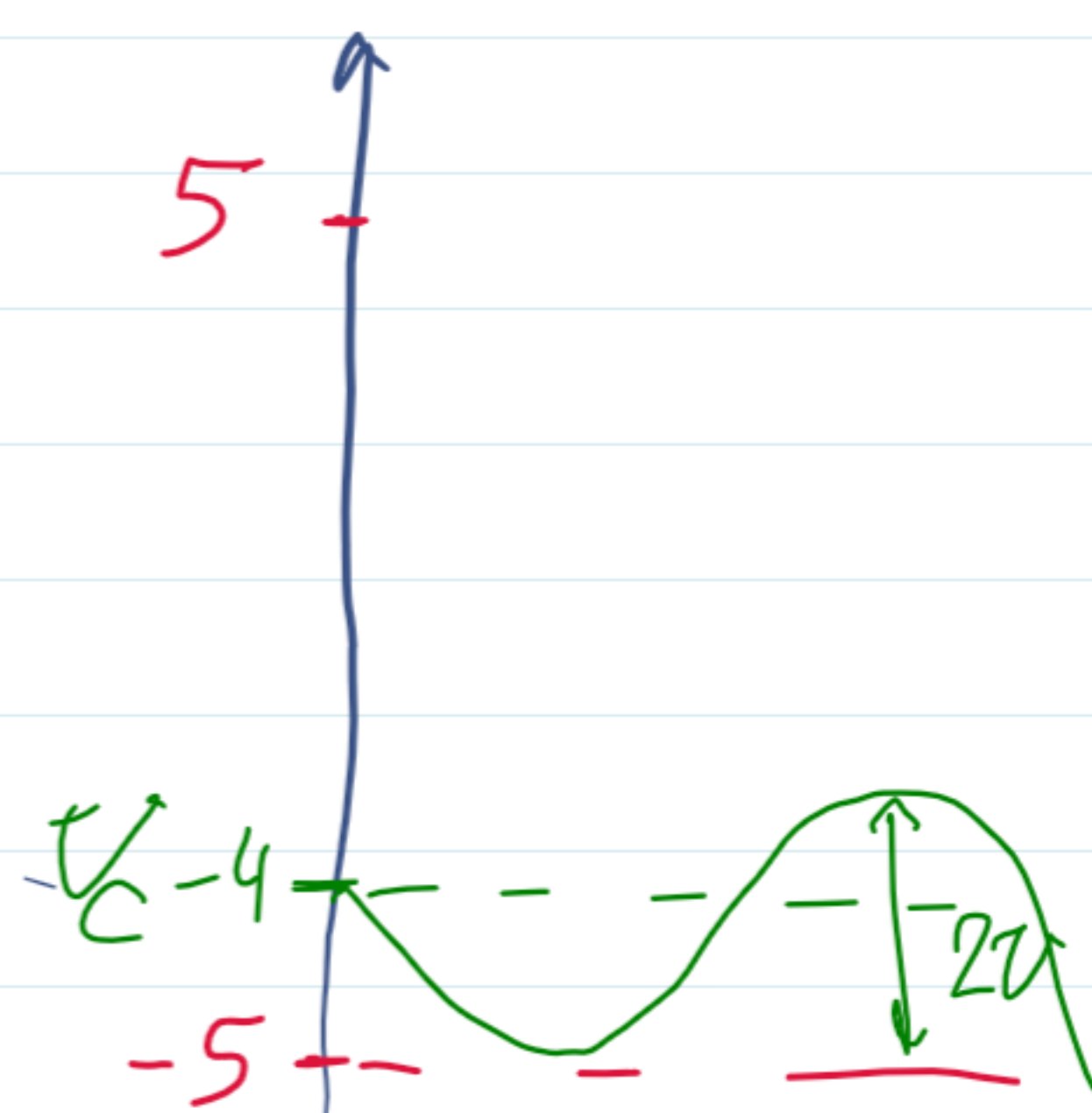
$$V_{EC} > |V_{CE,sat}|$$

$$> 0.2$$



$$R_L = 4k\Omega$$

$$V_C = -4$$



اگر C_E وجود نداشته باشد، بهره به شدت افت پیدا می کند. چراغ تقویت به دلتائی که بررسی r_{π} می باشد وابسته است. وجود R_E باعث می شود بخش زیادی از این ولتاژ بررسی R_E بیفتد. مثل تقسیم ولتاژ r_{π} و βR_E است.